

IF2240 – Basis Data Database Development Process

SEMESTER II TAHUN AJARAN 2025/2026



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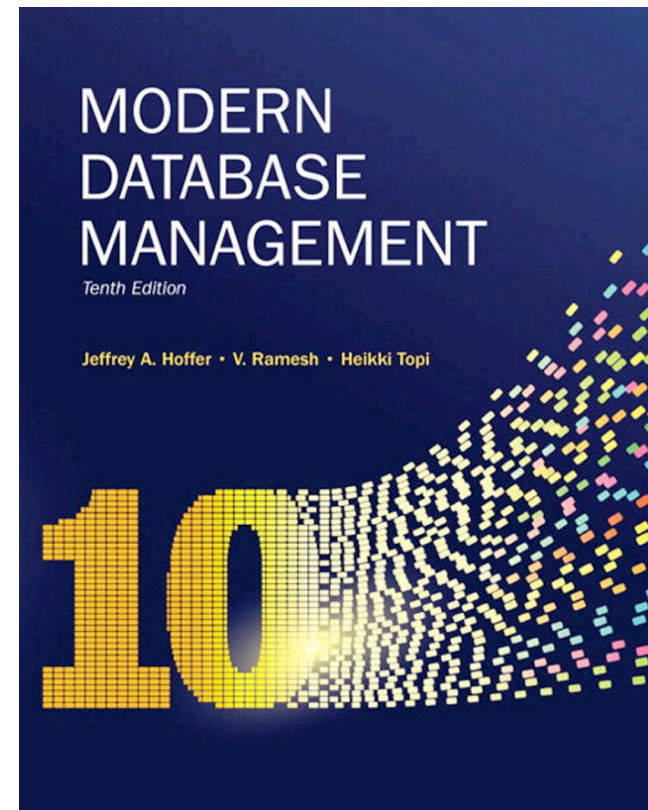
Sumber

Jeffrey A. Hoffer, V. Ramesh, Heikki Topi: "Modern Database Management", 10th Edition

- **Chapter 1:** The Database Environment and Development Process



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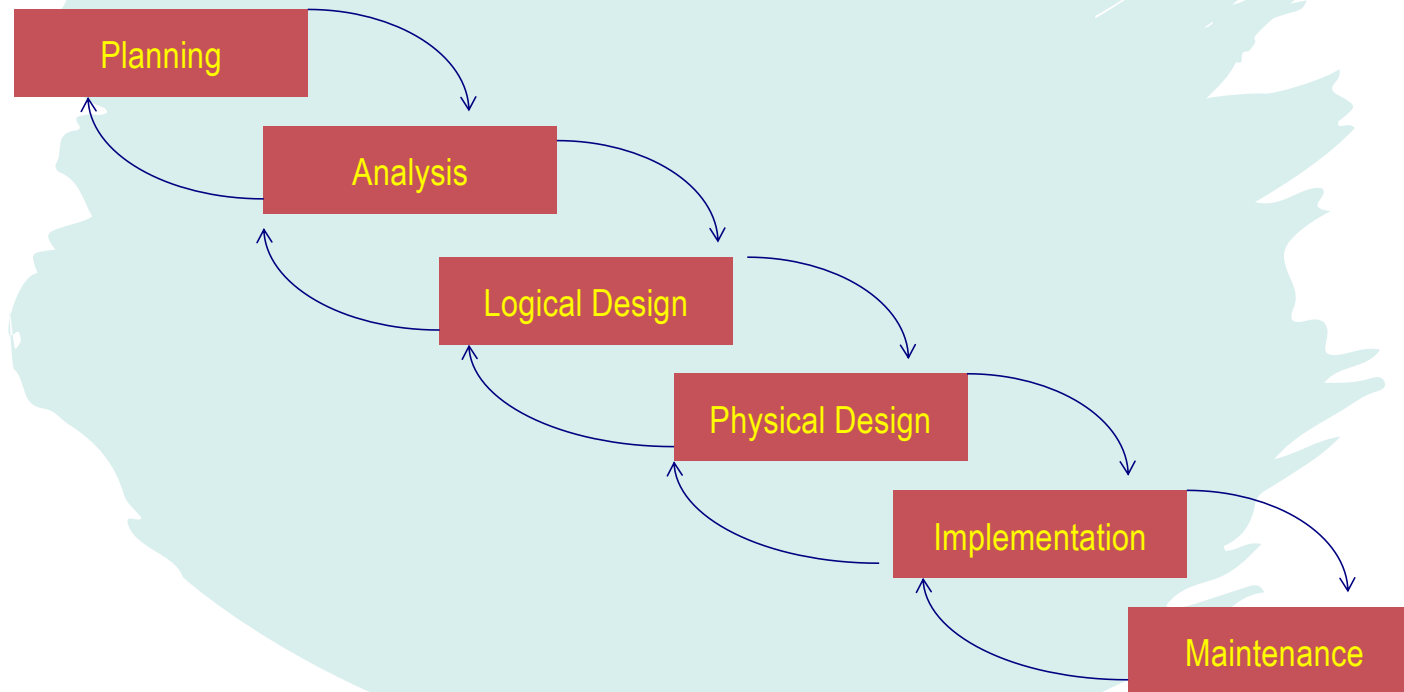
Two Approaches to Database and IS Development

- SDLC
 - System Development Life Cycle
 - Detailed, well-planned development process
 - Time-consuming, but comprehensive
 - Long development cycle
- Prototyping
 - Rapid application development (RAD)
 - cursory attempt at conceptual data modeling
 - Define database during development of initial prototype
 - Repeat implementation and maintenance activities with new prototype versions



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Systems Development Life Cycle

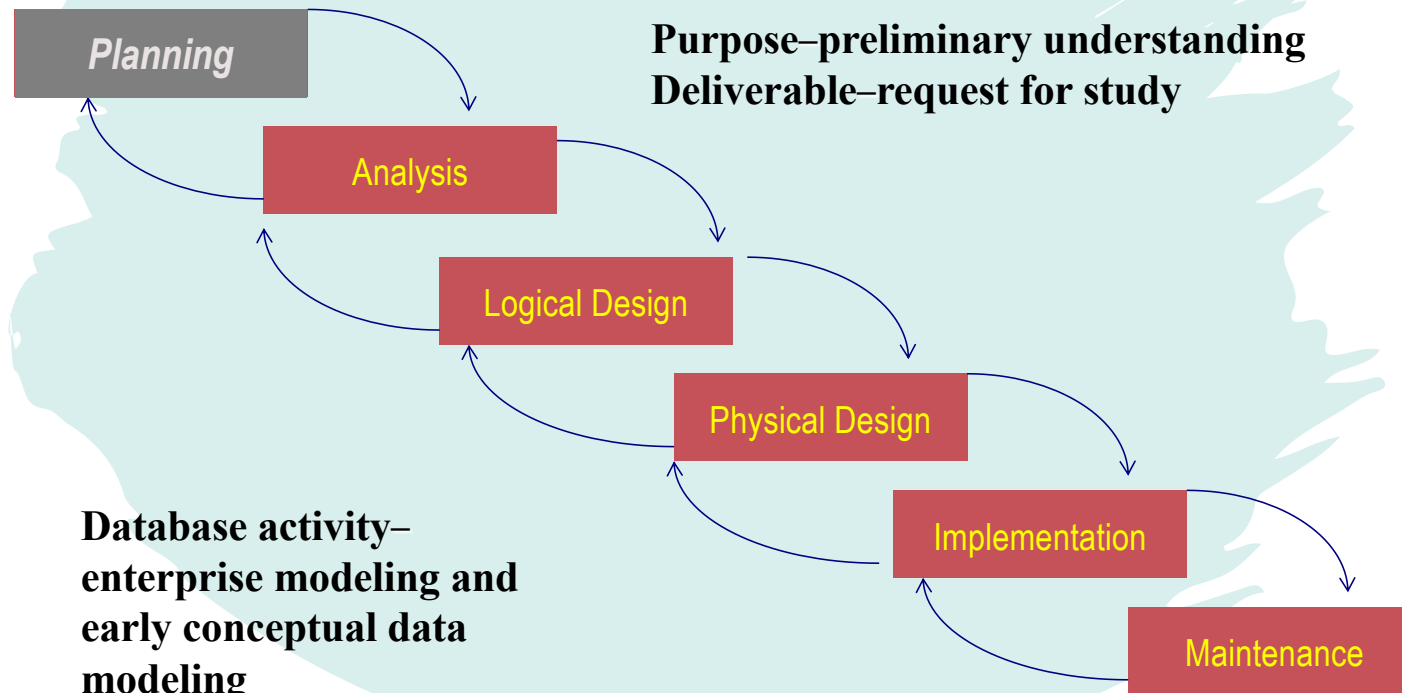


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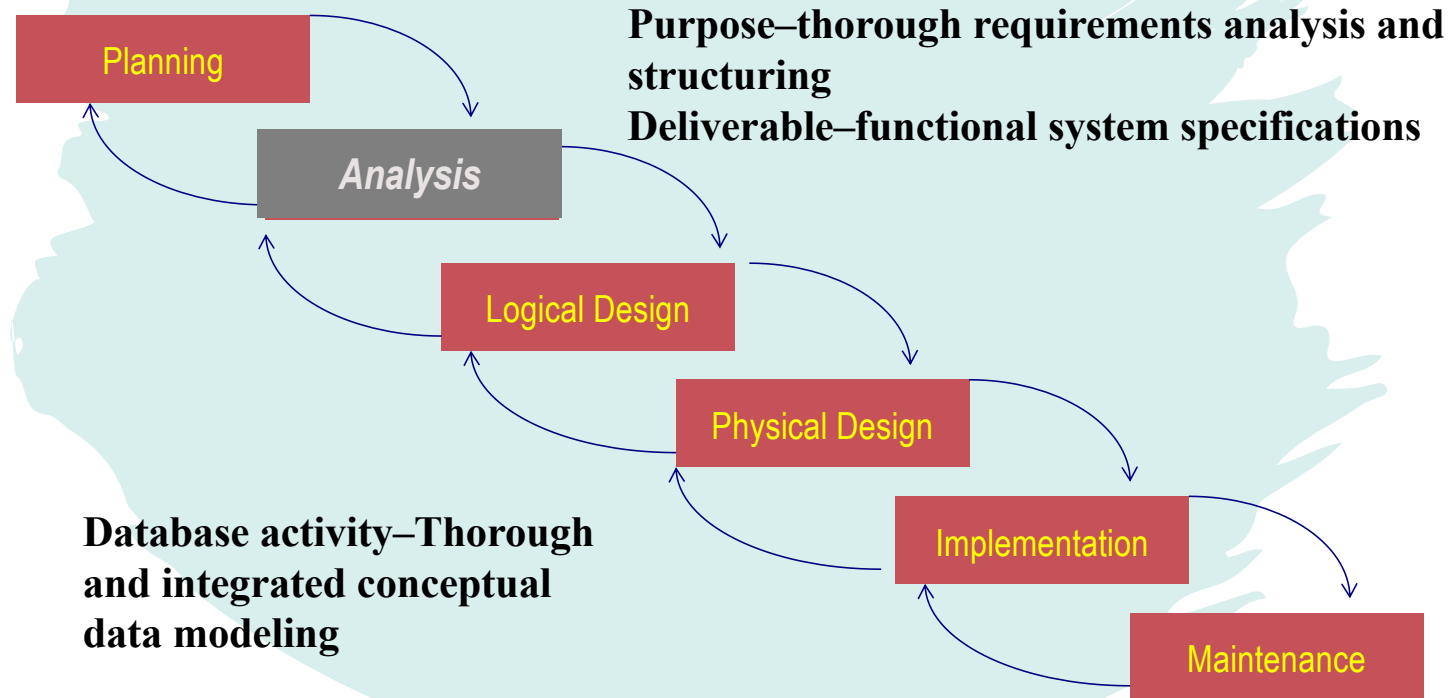
Systems Development Life Cycle (cont.)



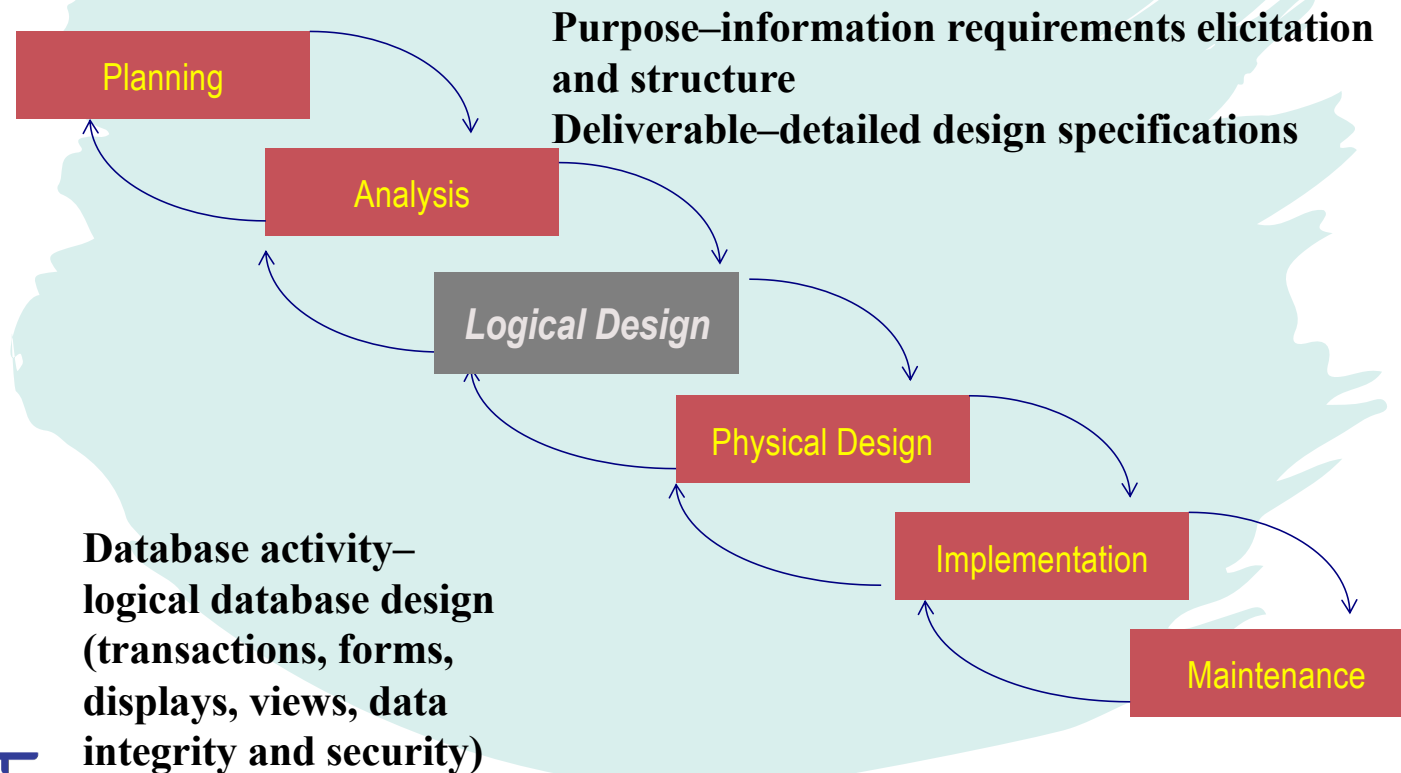
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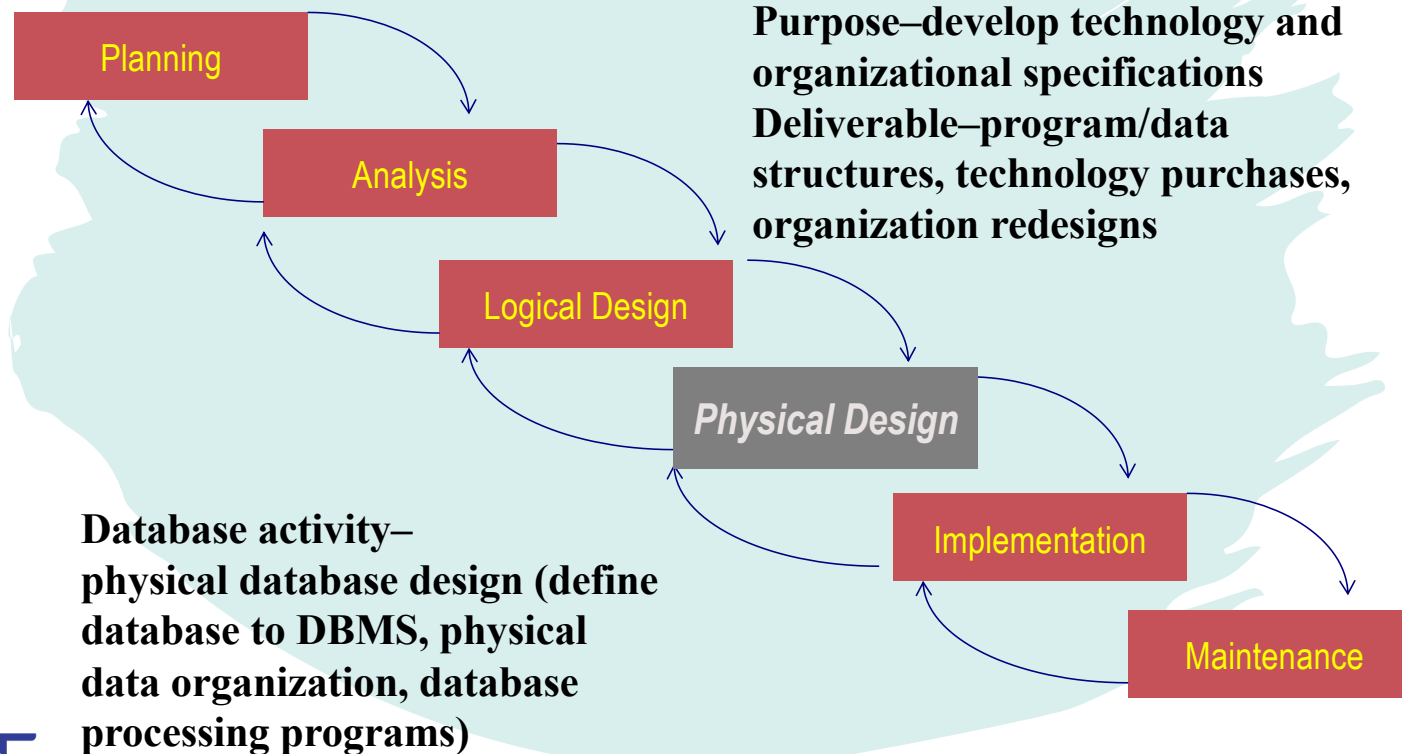
Systems Development Life Cycle (cont.)



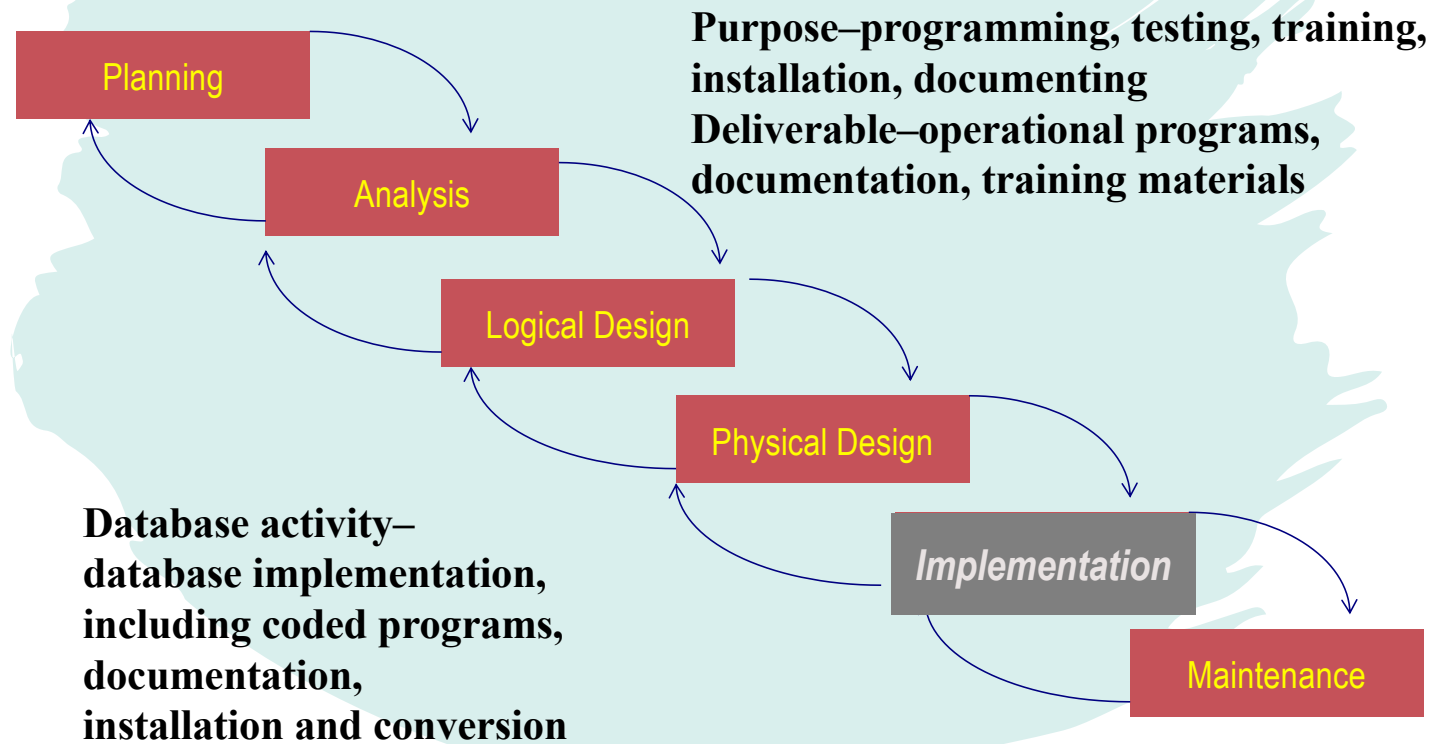
Systems Development Life Cycle (cont.)



Systems Development Life Cycle (cont.)



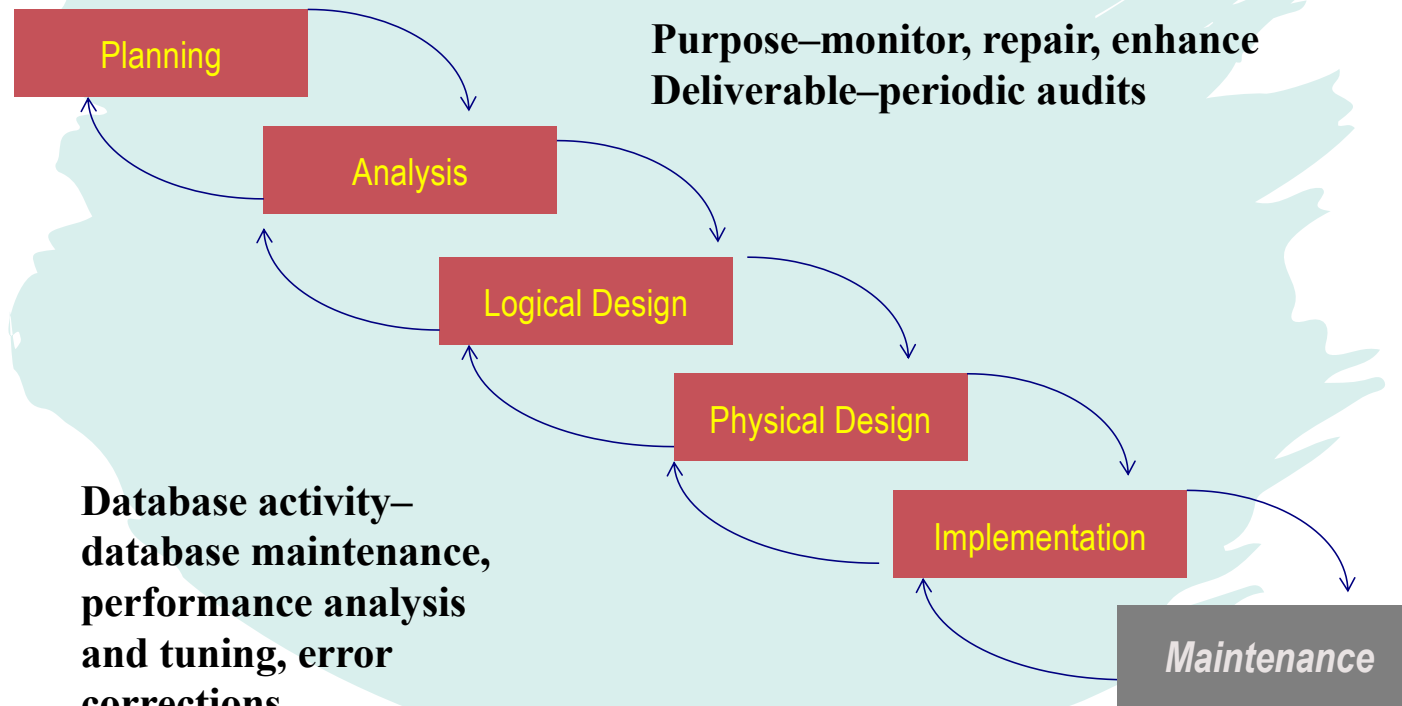
Systems Development Life Cycle (cont.)



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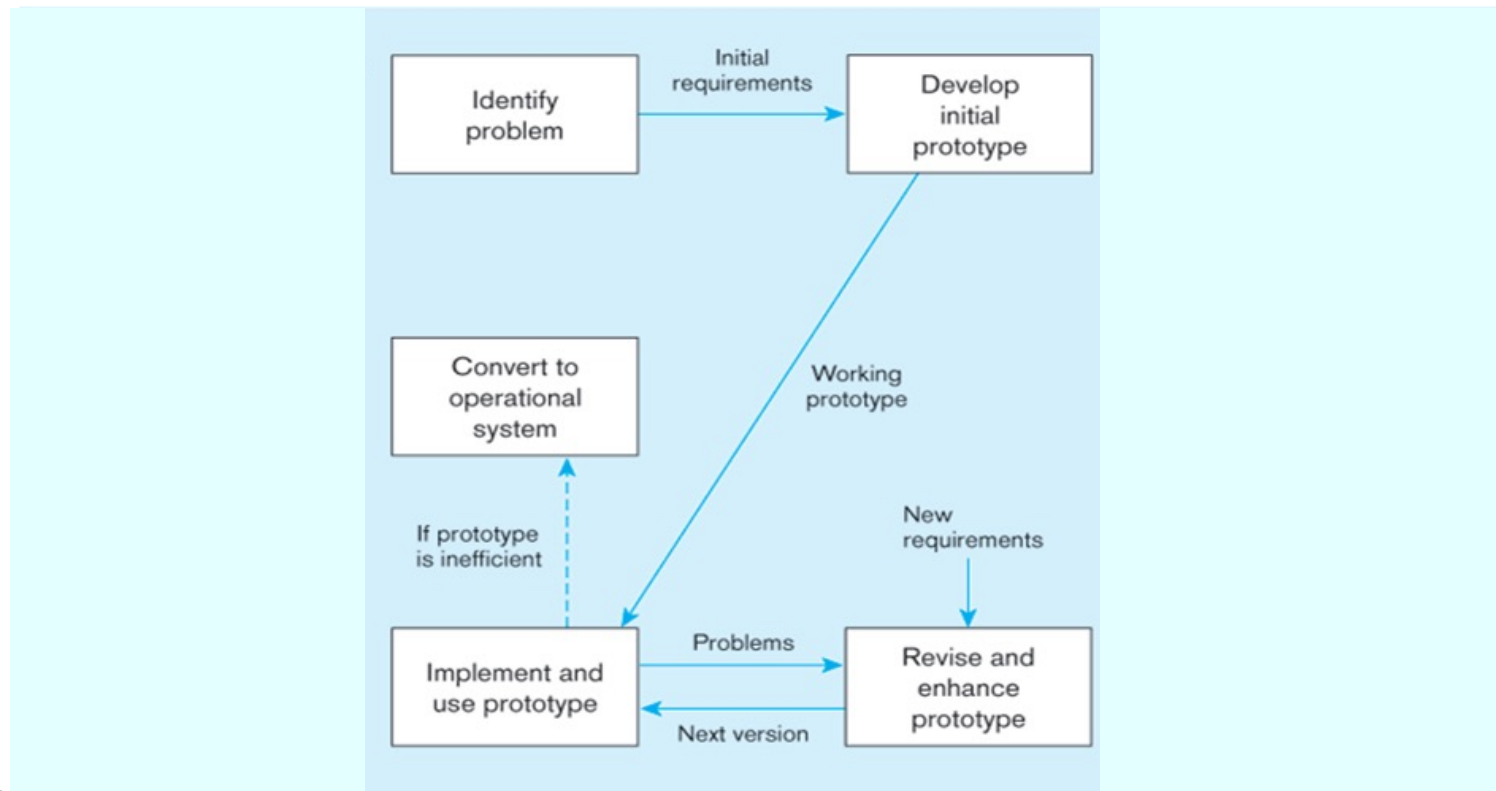
Systems Development Life Cycle (cont.)



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Prototyping Database Methodology

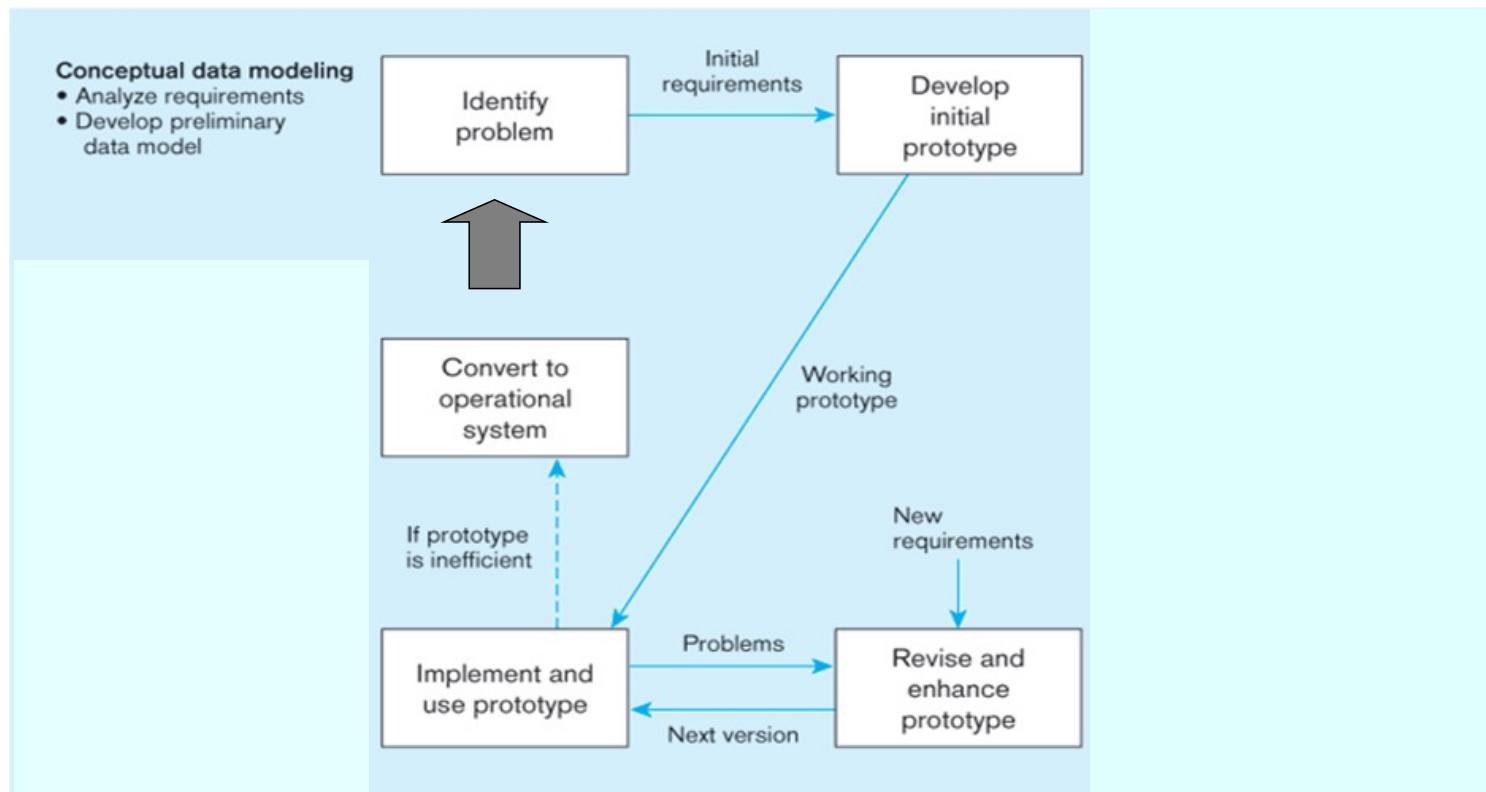


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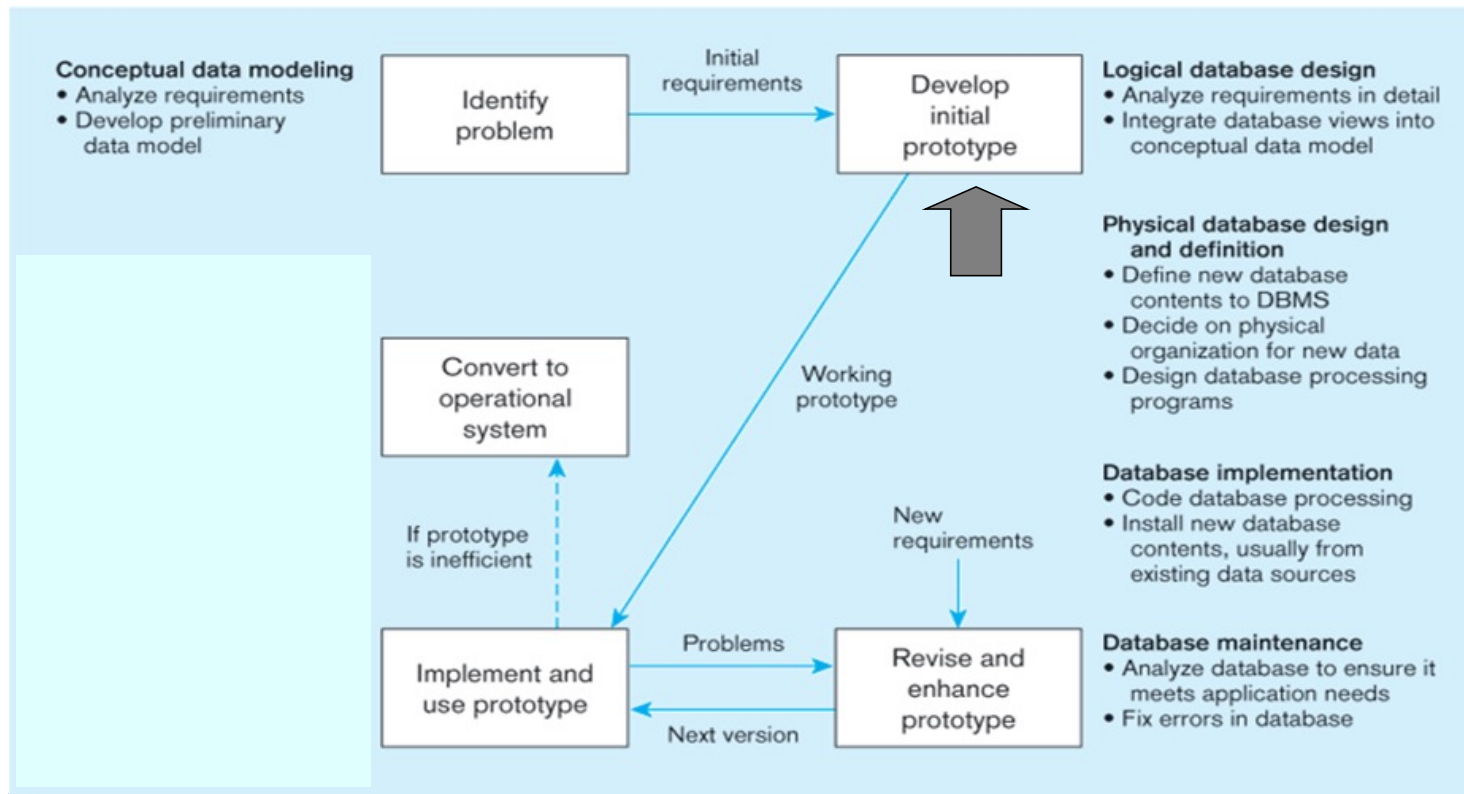
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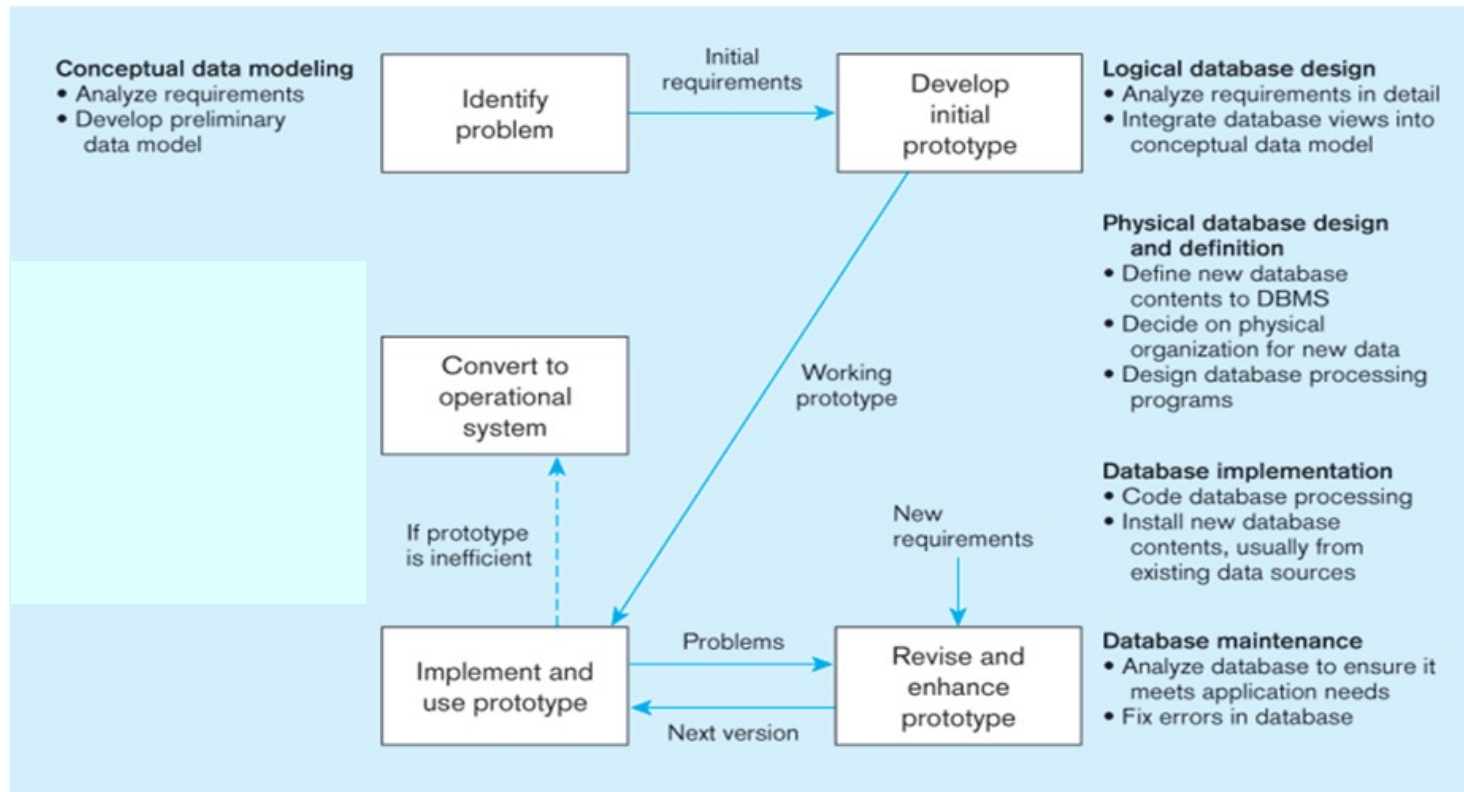
Prototyping Database Methodology (cont.)



Prototyping Database Methodology (cont.)



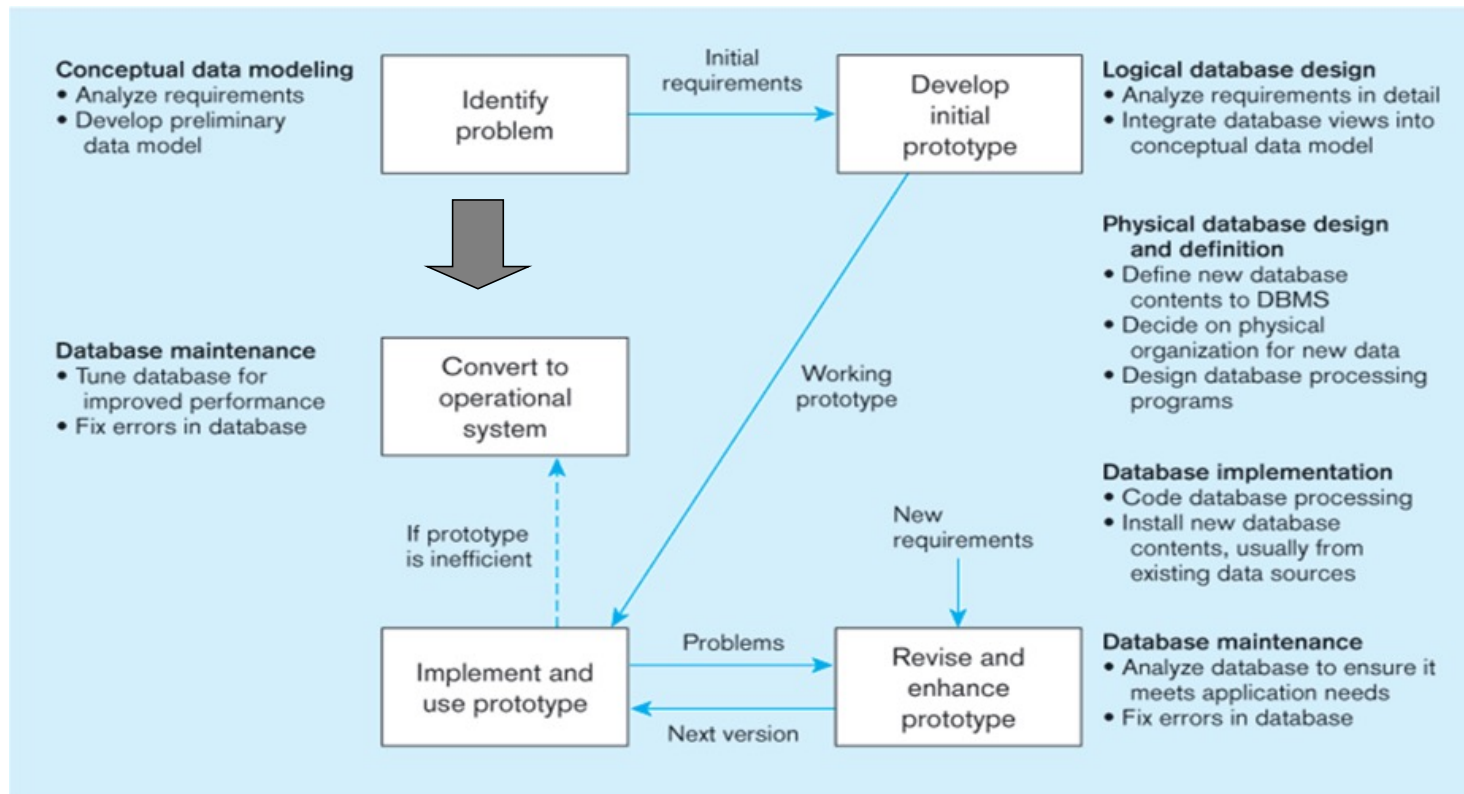
Prototyping Database Methodology (cont.)



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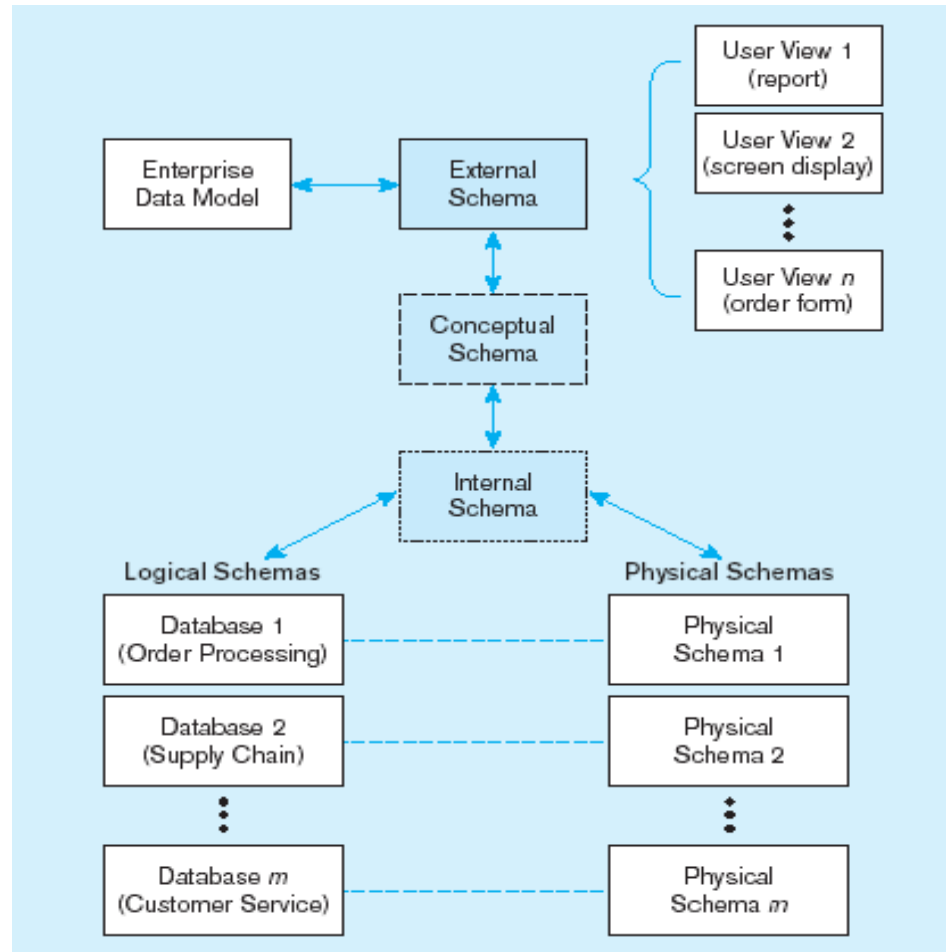
Prototyping Database Methodology (cont.)



Three-schema architecture

Different people have different views of the database...these are the external schema

The internal schema is the underlying design and implementation

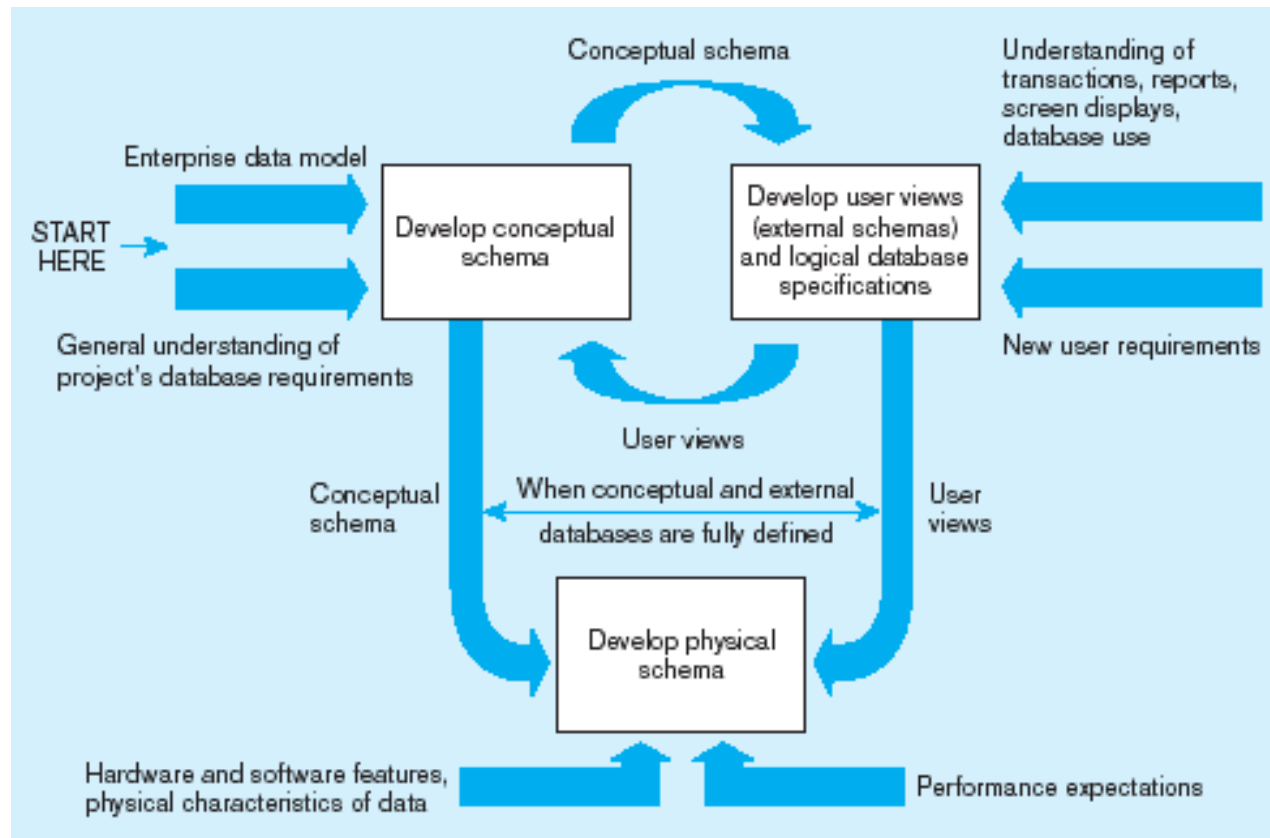


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Developing the three-tiered architecture



Three-tiered client/server database architecture

